

REPORT

Expense Tracker Using Python

Internship Program

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Technology Used: Python Programming

Introduction

Expense management is an important aspect of personal finance and budgeting. Keeping track of daily expenses helps individuals understand their spending habits and make better financial decisions. This project was developed using Python to demonstrate fundamental programming concepts such as variables, mathematical operations, loops, conditional statements, and user input handling.

The Expense Tracker application allows users to enter multiple expense amounts and automatically calculates the total expenditure. The project introduces the concept of accumulators, where numerical values are continuously added and processed within a program.

Objective

The primary objectives of this project are:

- To understand accumulator variables in Python.
- To perform mathematical calculations using user input.
- To learn how to process continuous data entry.
- To implement loops and conditional statements.
- To build a simple expense management system.

Problem Statement

Tracking expenses manually can be time-consuming and prone to errors. The objective of this project is to create a digital expense tracker where users can:

- Enter multiple expense amounts.
- Calculate the total amount spent.
- View the final expense summary.
- Exit the application when desired.

The program should continuously accept expense entries and update the total expenditure dynamically.

System Requirements

Hardware Requirements

- Computer or Laptop
- Minimum 2 GB RAM

Software Requirements

- Python 3.x
- Visual Studio Code / PyCharm / IDLE

Methodology

The project follows the IPO (Input-Process-Output) model.

Input

- User enters expense amounts.
- User chooses when to stop entering data.

Process

- Expenses are accepted one by one.
- Each expense is added to a running total using an accumulator variable.
- The program continues processing until the user exits.

Output

- Displays the total amount spent.
- Shows the final expense summary.

Algorithm

1. Initialize a variable named total with value 0.
2. Accept expense input from the user.
3. Add the entered expense to the total amount.
4. Repeat the process for additional expenses.
5. Allow the user to stop entering expenses.
6. Display the final total expenditure.
7. Terminate the program.

Source Code

```
total = 0

while True:
    expense = input("Enter expense amount (or 'done' to finish): ")

    if expense.lower() == "done":
        break

    total += float(expense)

print("\n----- Expense Summary -----")
print("Total Spent: ₹", total)
```

Output Screenshots

```
Enter expense amount (or 'done' to finish): 100
Enter expense amount (or 'done' to finish): 50
Enter expense amount (or 'done' to finish): 20
Enter expense amount (or 'done' to finish): done

----- Expense Summary -----
Total Spent: ₹ 170.0
```

Features

- Simple and user-friendly interface.
- Automatic calculation of total expenses.
- Continuous data entry using loops.
- Real-time expense accumulation.
- Easy to understand and modify.

Learning Outcomes

Through this project, the following concepts were learned:

- Variables and Data Types
- Mathematical Operations
- Accumulator Pattern
- Loops (while loop)
- Conditional Statements (if statement)
- User Input Handling
- Basic Data Processing

Future Enhancements

The application can be improved by adding:

- Expense categories (Food, Travel, Shopping, etc.).
- Daily and monthly expense reports.
- Expense deletion functionality.
- File storage for saving expense history.
- Graphical User Interface (GUI).
- Database integration using SQLite or MySQL.
- Expense analytics and visualization charts.

Conclusion

The Expense Tracker Application successfully demonstrates the use of Python programming fundamentals to process and manage financial data. The project provides practical experience in handling user input, performing calculations, and implementing accumulator logic. It serves as a strong foundation for learning advanced concepts such as file handling, databases, financial applications, and backend development.

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